
Experiment No: 1

Date:

THREE NODE POINT TO POINT NETWORK

Aim: *Implement three nodes point – to – point network with duplex links between them. Set the queue size, vary the bandwidth and find the number of packets dropped.*

```
set ns [new Simulator]           # Letter S is capital
set nf [open lab1.nam w]         # open a nam trace file in write mode
$ns namtrace-all $nf           # nf  nam filename
set tf [open lab1.tr w]         # tf  trace filename
$ns trace-all $tf

proc finish { } {
    global ns nf tf
    $ns flush-trace              # clears trace file contents
    close $nf
    close $tf
    exec nam lab1.nam &
    exit 0
}

set n0 [$ns node]               # creates 3 nodes
set n2 [$ns node]
set n3 [$ns node]

$ns duplex-link $n0 $n2 200Mb 10ms DropTail # establishing links
$ns duplex-link $n2 $n3 1Mb 1000ms DropTail
$ns queue-limit $n0 $n2 10

set udp0 [new Agent/UDP]        # attaching transport layer protocols
$ns attach-agent $n0 $udp0
set cbr0 [new Application/Traffic/CBR] # attaching application layer protocols
$cbr0 set packetSize_ 500
$cbr0 set interval_ 0.005
$cbr0 attach-agent $udp0

set null0 [new Agent/Null]      # creating sink(destination) node
$ns attach-agent $n3 $null0
$ns connect $udp0 $null0

$ns at 0.1 "$cbr0 start"
$ns at 1.0 "finish"
$ns run
```

AWK file: *(Open a new editor using “vi command” and write awk file and save with “.awk” extension)*

#immediately after BEGIN should open braces ‘{‘

```

BEGIN{ c=0;}
{
  if($1=="d")
  {
    c++;
    printf("%s\t%s\n", $5, $11);
  }
}
END{ printf("The number of packets dropped is %d\n", c); }

```

Steps for execution

- ❑ Open gedit editor and type program. Program name should have the extension “.tcl ”

[root@localhost ~]# gedit lab1.tcl

- ❑ Save the program and close the file.
- ❑ Open gedit editor and type **awk** program. Program name should have the extension “.awk ”

[root@localhost ~]# gedit lab1.awk

- ❑ Save the program and close the file.
- ❑ Run the simulation program

[root@localhost~]# ns lab1.tcl

- ❑ Here “ns” indicates network simulator. We get the topology shown in the snapshot.
- ❑ Now press the play button in the simulation window and the simulation will begins.
- ❑ After simulation is completed run **awk file** to see the output ,

[root@localhost~]# awk -f lab1.awk lab1.tr

- ❑ To see the trace file contents open the file as ,

[root@localhost~]# gedit lab1.tr

Trace file contains 12 columns:

Event type, Event time, From Node, To Node, Packet Type, Packet Size, Flags (indicated by -----), Flow ID, Source address, Destination address, Sequence ID, Packet ID

The image displays three screenshots from a Linux terminal environment. The leftmost screenshot shows the contents of the trace file 'lab1.tr', which is a table with 12 columns: Event type, Event time, From Node, To Node, Packet Type, Packet Size, Flags, Flow ID, Source address, Destination address, Sequence ID, and Packet ID. The middle screenshot shows a network topology diagram with three nodes (0, 1, 2) and their connections. The rightmost screenshot shows the output of the command 'awk -f PA1.awk lab01.tr', which lists the number of packets dropped for each flow ID (139 to 182) and a summary line indicating that the total number of packets dropped is 16.

Contents of Trace File

Topology

Output

Experiment No: 2

Date:

TRANSMISSION OF PING MESSAGE

Aim: *Implement transmission of ping messages/trace route over a network topology consisting of 6 nodes and find the number of packets dropped due to congestion.*

set ns [new Simulator]

set nf [open lab2.nam w]
\$ns namtrace-all \$nf

set tf [open lab2.tr w]
\$ns trace-all \$tf

set n0 [\$ns node]
set n1 [\$ns node]
set n2 [\$ns node]
set n3 [\$ns node]
set n4 [\$ns node]
set n5 [\$ns node]

\$ns duplex-link \$n0 \$n4 1005Mb 1ms DropTail
\$ns duplex-link \$n1 \$n4 50Mb 1ms DropTail
\$ns duplex-link \$n2 \$n4 2000Mb 1ms DropTail
\$ns duplex-link \$n3 \$n4 200Mb 1ms DropTail
\$ns duplex-link \$n4 \$n5 1Mb 1ms DropTail

set p1 [new Agent/Ping] # letters A and P should be capital
\$ns attach-agent \$n0 \$p1
\$p1 set packetSize_ 50000
\$p1 set interval_ 0.0001

set p2 [new Agent/Ping] # letters A and P should be capital
\$ns attach-agent \$n1 \$p2

set p3 [new Agent/Ping] # letters A and P should be capital
\$ns attach-agent \$n2 \$p3
\$p3 set packetSize_ 30000
\$p3 set interval_ 0.00001

set p4 [new Agent/Ping] # letters A and P should be capital
\$ns attach-agent \$n3 \$p4

set p5 [new Agent/Ping] # letters A and P should be capital
\$ns attach-agent \$n5 \$p5

\$ns queue-limit \$n0 \$n4 5
\$ns queue-limit \$n2 \$n4 3

```
$ns queue-limit $n4 $n5 2
Agent/Ping instproc rcv {from rtt} {
$self instvar node_
puts "node [$node_ id] received answer from $from with round trip time $rtt msec"
}
# please provide space between $node_ and id. No space between $ and from. No space
between and $ and rtt */
```

```
$ns connect $p1 $p5
$ns connect $p3 $p4
```

```
proc finish { } {
global ns nf tf
$ns flush-trace
close $nf
close $tf
exec nam lab2.nam &
exit 0
}
```

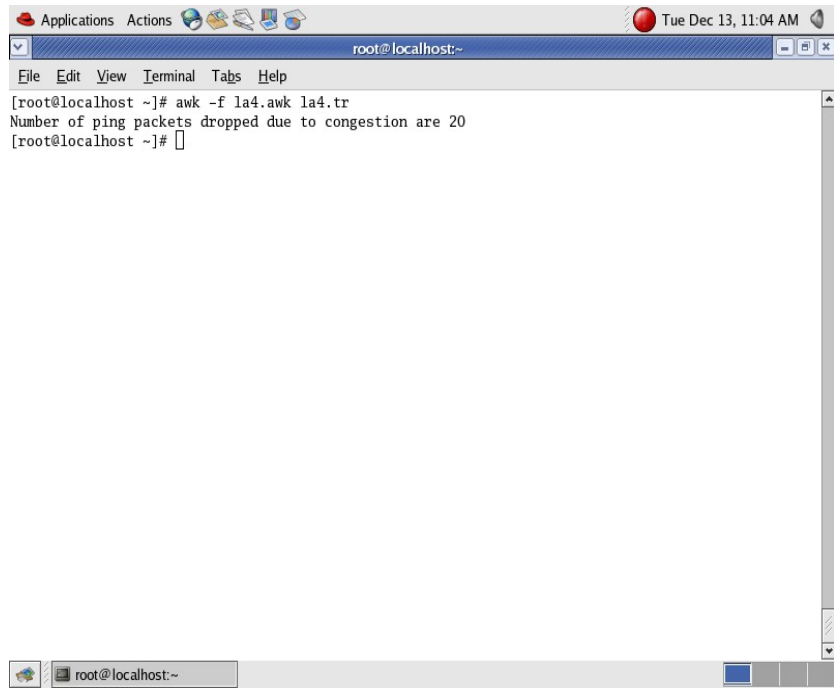
```
$ns at 0.1 "$p1 send"
$ns at 0.2 "$p1 send"
$ns at 0.3 "$p1 send"
$ns at 0.4 "$p1 send"
$ns at 0.5 "$p1 send"
$ns at 0.6 "$p1 send"
$ns at 0.7 "$p1 send"
$ns at 0.8 "$p1 send"
$ns at 0.9 "$p1 send"
$ns at 1.0 "$p1 send"
```

```
$ns at 0.1 "$p3 send"
$ns at 0.2 "$p3 send"
$ns at 0.3 "$p3 send"
$ns at 0.4 "$p3 send"
$ns at 0.5 "$p3 send"
$ns at 0.6 "$p3 send"
$ns at 0.7 "$p3 send"
$ns at 0.8 "$p3 send"
$ns at 0.9 "$p3 send"
$ns at 1.0 "$p3 send"
```

```
$ns at 2.0 "finish"
$ns run
```

AWK file: (Open a new editor using “gedit command” and write awk file and save with “.awk” extension)

```
BEGIN{
drop=0;
}
```



The image shows a terminal window titled "root@localhost:~". The window has a menu bar with "File", "Edit", "View", "Terminal", "Tags", and "Help". The terminal content shows the command `awk -f la4.awk la4.tr` being executed, which outputs the text "Number of ping packets dropped due to congestion are 20". The prompt `[root@localhost ~]#` is visible at the end of the line. The window's title bar includes "Applications", "Actions", and a clock showing "Tue Dec 13, 11:04 AM".

```
[root@localhost ~]# awk -f la4.awk la4.tr
Number of ping packets dropped due to congestion are 20
[root@localhost ~]#
```

Output